

Design and Development of a Low-Cost Lightweight Tea Leaf Collection Bucket

Team Ergonomics

E/21/037 | E/21/094 | E/21/365

1. Motivation

The motivation for this project stems from the need to address the severe physical strain and inefficiencies associated with traditional tea harvesting, which is a highly labor-intensive activity in tea plantations. Currently, workers use woven bamboo or cane baskets that cause significant worker fatigue, back pain, and neck strain due to heavy empty weights and poor load distribution. Furthermore, existing baskets lack proper ventilation, leading to heat development that causes fermentation and permutation of the tea leaves. We aim to design an optimized, low-cost solution that improves harvesting productivity and drastically enhances worker comfort.

2. Overview

2.1 Significance of the project

This project carries significant "Big Three" impacts across economic, social, and quality domains. Socially, lowering physical strain eliminates chronic shoulder pain and fatigue, improving the overall well-being and resilience of the plantation working communities. Environmentally and in terms of quality, the design promotes sustainable harvesting by drastically reducing crop waste caused by heat-induced permutation of bottom leaves. Economically, by facilitating a larger capacity of 10-11 kg per trip at a lower manufacturing cost, the solution provides massive economic relief and higher daily productivity to plantation owners at a fleet scale.

2.2 Description of the Project

We are designing a wearable architecture that replaces heavy woven bamboo/cane to reduce empty weight and improve worker mobility. Our task is to develop the system to:

- Integrate shoulder strap structural supports to eliminate localized worker strain.
- Provide an engineered ventilated cavity to prevent internal heat buildup.
- Apply rigid support bars to maintain structural integrity and prevent the crushing of bottom leaves.

2.3 Background of the Project

Traditionally, tea leaves are collected using woven baskets carried on the back with a strap around the head or shoulders, which typically weigh 2.5–3 kg empty and 10–11 kg full. Several alternatives exist, such as nylon collection bags, which unfortunately limit the maximum collectable weight to only 6–7 kg and cause leaves at the bottom to get crushed. Other existing ergonomic baskets have a total cost of Rs. 1275, which limits widespread adoption. Many of these alternative systems are either too expensive or unsuitable for steep plantation terrain.

3. Methodology

3.1 Design Phase

The proposed solution replaces the traditional heavy basket with a lightweight synthetic material bucket featuring specialized support padding and structural straps in the shoulder area. To solve thermodynamic issues, the design incorporates internal support bars to reduce pressure development alongside targeted ventilation slots. The design process strictly targets a reduced manufacturing cost of Rs. 1000 to ensure the purchase cost can be quickly recovered through productivity improvements.

3.2 Implementation Phase

The execution of the design involves:

- Conducting an ergonomic baseline study using existing traditional baskets to identify specific pain points.
- Selecting low-cost, lightweight synthetic materials to achieve the Rs. 1000 target.
- Creating a CAD model of the internal support frame and bucket frame.

3.3 Testing Phase

The testing phase focuses on evaluating key components through software simulation and dominant load-bearing analysis. The main aspects to be tested include:

- Thermal flow analysis to ensure heat dissipation.
- Structural stress testing to verify the durability of the support bars.

3.4 Evaluation Phase

- Conduct field visualization and capacity testing to confirm the bucket can safely hold the 11kg limit.
- Verify that the design results in zero leaf permutation or crushing.

4. Features

- **Wearable Architecture:** Reduces empty weight and improves mobility.
- **Ventilated Cavity & Support Bars:** Prevents internal heat buildup, fermentation, and crushing of bottom leaves.
- **Load Distribution:** Integrated shoulder straps provide anatomical support, eliminating localized strain.
- **Increased Capacity:** Safely handles a maximum collectable weight of 10-11 kg.
- **Economical:** Reduced target cost of Rs. 1000 per unit.

5. Project Planning

The project is scheduled across a 4-month timeline from May to August.

- **May:** Research & Conceptualization, including concept finalization.
- **June:** Detailed Design & Analysis.
- **July:** Simulation, Prototyping & Verification (Fabrication & Bench testing).
- **August:** Final Evaluation & Documentation (Integration in the field, Main experimental campaign, and Final reporting).

6. Hardware and Software Requirements

Hardware / Materials:

- Low-cost, lightweight synthetic materials.
- Rigid internal support bars.
- Specialized support padding and straps.

Software Tools:

- **SolidWorks:** For CAD design of the bucket frame and ventilation slots.
- **ANSYS:** For thermal flow analysis and structural stress testing.